



Research Paper

Comprehensive industry-relevant black soldier fly bioconversion characterisation by a novel chamber system

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ABSTRACT

Black soldier fly larvae (BSFL) efficiently convert biowaste into valuable animal feed. Sustainable and reliable bioconversion is desirable to achieve optimal economic and environmental outcomes. Thus, science and industry require an accessible research platform to study complex bioconversion processes under conditions mirroring industrial-scale settings. In this study, industry-relevant respiration chambers were designed, tested, and replicated for BSFL feeding trials. Each open-circuit chamber housed three industrial rearing crates. The substrate/frass and air temperature, mass change, NH₃ and CO₂ emissions, and relative humidity were measured. The design was validated for CO₂ recovery, airtightness, airflow homogeneity, and BSFL performance using firstly, a uniform control substrate and secondly, uniform food waste across four parallel chambers. In a third trial, the composition of food waste was varied across parallel chambers to detect differences in metabolic processes. For trials using uniform substrates, low variability across chambers in performance parameters confirmed the reproducibility and comparability of the design (e.g. bioconversion rate: <1%, final larval mass: ≤2 mg, standard deviations, dry matter based). In contrast, the trial with varying food waste compositions showed a strong effect on average substrate/frass temperature (e.g. 31.5 °C vs 41.8 °C) and final dry larval mass (e.g. 67 mg vs 40 mg). This is the first study to systematically assess heat generation directly from heterogeneous food waste, a crucial parameter for efficient BSFL bioconversion. These chambers provide an opportunity for science and industry to thoroughly assess and understand the metabolic bioconversion characteristics. The findings are key for the optimisation of sustainable bioconversion processes.

1. Introduction

Globally, almost one-quarter of the calories produced for human nutrition are lost along the food chain and never consumed (Searchinger et al., 2019). Next to saving resources, waste recycling within the food system can enhance food security by preventing nutrient loss in landfills (Gunders and Bloom, 2017; Irani et al., 2018; Wang et al., 2021). Food supply chains are vulnerable, especially to evolving geopolitical and climate change-related challenges (Ghadge et al., 2020). Larger urban areas with increasing populations, space constraints, and limited self-sufficiency in food production are looking to decrease their sensitivity to supply disturbances by incorporating food production pipelines

(Gulyas and Edmondson, 2021). Simultaneously, the wealth of urban areas results in higher biowaste generation and consumption of animal-based food (Milford et al., 2019; Parfitt et al., 2010). From environmental and food security perspectives, cities can benefit from a circular food system. This requires approaches that prioritise nutrient recovery in a manner that can efficiently upcycle biowaste and provide agricultural input.

The black soldier fly larva (BSFL, *Hermetia illucens*) is an emerging tool for biowaste recycling. It can feed on a broad range of biowastes (e.g. manure, food industry by-products, municipal food waste, and slaughterhouse waste) and convert the nutrients in the biowaste into larval biomass (Banks et al., 2014; Gebremikael et al., 2020; Newton

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et al., 2005), which has myriad applications. For instance, the dry larval biomass can contain 20 %–60 % protein and more than 25 % lipids, which can be captured as valuable animal feed (Gold et al., 2018; Kim et al., 2020; Siegrist et al., 2023) with potential animal health benefits (Huis, 2020). When reared on underutilised biowaste streams, such as food waste, BSFL can be more sustainable than current feed ingredients, such as fishmeal (Smetana et al., 2019). Additionally, marketable substances, such as chitin or antimicrobial proteins, can be extracted from the larvae (Ravi et al., 2020). Finally, the residues from the rearing process (hereafter referred to as ‘frass’) are plant fertilisers (Schmitt and de Vries, 2020). Due to their versatility BSFL become attractive for urban biowaste recycling, which is intrinsically heterogeneous. Such heterogeneous waste often exhibits significant variations in its properties, including its moisture content and its nutrient composition (Abdel-Shafy and Mansour, 2018). This variability can lead to common challenges in BSFL bioconversion, such as high heat production (Erbländ et al., 2021), insufficient frass drying (Lalander et al., 2020) and high NH_3 and CO_2 emissions (Froonickx et al., 2023; Parodi et al., 2021). Even though these processes are interrelated, they are not studied in combination. Their elucidation is key since they can have dramatic consequences for the efficiency and reliability of the bioconversion. For instance, heat production can influence larval escape, growth and mortality as well as the ability to separate frass and larvae, which are operational problems that undermine the sustained economic and environmental viability of this technology.

The bioconversion process has mostly been characterised by monitoring larval mass, biowaste reduction, bioconversion, and survival rates in controlled feeding trials. Because BSFL live in their substrate/frass (i. e. biowaste), the temperature can be significantly different from the ambient temperature. Thus, the importance of substrate/frass temperature monitoring has been recognised recently but it has been monitored by only a few studies (Erbländ et al., 2021; Li et al., 2023). In fact, Erbländ et al. (2021) prototyped a ventilated incubator (six crates, 0.27 m^2 per crate) that tracked mass loss and substrate/frass temperature throughout BSFL bioconversion with chicken feed. The data suggest that microbial activity plays an important role in the rearing of BSFL. Nevertheless, there is a lack of research that attempts to reconstruct a more comprehensive picture of BSFL bioconversion by measuring both insect life history parameters and the abiotic environment by monitoring temperature fluctuations as well as the exchange of water and emissions during the rearing process.

From answering fundamental biological questions to designing climate systems for efficient BSFL bioconversion, the simultaneous quantification of gaseous emissions and heat production during the bioconversion is key. Respiration chambers are commonly used to monitor gaseous emissions from living organisms to determine energy and nutrient requirements and collect data for environmental impact assessments of protein production. For instance respiration chambers are used for the assessment of the metabolic activity of plants, soil, or large livestock, such as cattle, however chamber size and the instruments used vary significantly (Albrizio and Steduto, 2003; Carbone et al., 2019; Hammond et al., 2016; Heetkamp et al., 2015). Recent evaluations of BSFL bioconversion have been conducted using dynamic open-circuit respiration chambers, where gas concentrations were continuously measured in ventilated chambers throughout the process (Padmanabha and Streif, 2019; Parodi et al., 2020, 2021). Parodi et al. (2020, 2021) monitored BSFL bioconversion with pig manure and an industrial rearing substrate in an open-circuit respiration chamber (three rearing crates per chamber, 0.15 m^2 per crate or one rearing crate per chamber, 0.5 m^2 per crate). The goal was to track the development of CO_2 , N_2O , CH_4 , O_2 and NH_3 gas emissions and to calculate comprehensive nitrogen, carbon, and energy balances to assess the environmental sustainability of BSFL rearing. In contrast, Padmanabha and Streif (2019) designed a low-cost chamber for automated BSFL rearing and growth model development (Padmanabha et al., 2020, 2023). This setup was capable of monitoring CO_2 and O_2 concentrations, air

temperature, and humidity, as well as the substrate/frass temperature and humidity. However, it failed to capture NH_3 emissions, which are crucial indicators of larval protein metabolism and industrially relevant air pollutants (Froonickx et al., 2023). Additionally, only a single crate (0.12 m^2 per crate), smaller than industrial size, was housed in their setup. Consequently, there is a need for an accessible monitoring system that can comprehensively assess the metabolic activity and trends in the BSFL bioconversion processes of different substrates under conditions that resemble practical implementation scenarios while adhering to scientific standards.

Few studies have determined gaseous emissions (CO_2 , CH_4 , N_2O and NH_3) from BSFL food waste bioconversion, mainly for environmental impact assessments (Ermolaev et al., 2019; Lindberg et al., 2022; Mertenat et al., 2019; Pang et al., 2020). Large intervals between sampling points complicate the determination of total emissions and neglect common emission dynamics. Additionally, emission samples were drawn from static systems (non-ventilated containers at the time of sampling), in which increasing gas concentrations during measurement could affect instantaneous larval and microbial activities, thereby weakening the estimate. For instance, Guo et al. (2021) measured gaseous emissions from commercial BSFL food waste bioconversion. The sampling was performed at intervals of 4 h but only over a period of 3 days of the whole process, which did not enable the authors to draw conclusions for the whole process. Unlike composting, a 6–7-day BSFL bioconversion process is highly dynamic, and frequent sampling must cover the entire process duration. For instance, CO_2 emissions related to larval and microbial activity can peak within hours, while NH_3 emissions may surge only 2 days before the harvest of larvae (Coudron et al., 2024; Parodi et al., 2020, 2021). Heterogeneous food waste is a challenging but key substrate in urban biowaste management and sustainable BSFL rearing and, therefore, needs urgent further exploration.

The aim of this study was to propose, describe, and validate an industry-relevant bench-scale respiration system and use it for the first time for the high-resolution characterisation of BSFL food waste bioconversion. The system was validated for CO_2 recovery, air flow homogeneity, and its ability to capture the bioconversion within and across parallel chambers using food waste in comparison to a reference substrate. The results illustrated that a comprehensive BSFL bioconversion characterisation helps to understand BSFL food waste recycling and to identify areas for further research. Therefore, this study established a tool to improve the bioconversion performance of BSFL with urban biowaste for the production of feed and fertiliser for a circular food system.

2. Material and methods

2.1. Respiration chamber for fly larvae bioconversion

The respiration system included four equally designed respiration chambers that could be individually operated (Supplementary Fig. 1). Fig. 1 depicts the general setup and components of the respiration chamber, including the positions of sensors, whereas Table 1 depicts the key system specifications. The respiration chamber was ventilated through air inlet and outlet ducts for optimal sensor deployment and determination of emission production and water removal from the respiration system. The walls of the chamber were made of polyisocyanurate panels with an aluminium coating (30 mm; Kingspan, Kingscourt, Ireland) to resist humidity and corrosion. For optimal sampling and crate handling, the lid of the chamber was liftable from the top, additionally tightened by rubber bands at the rim to ensure the airtightness of the respiration system. The air inlet and outlet ducts were covered with a plastic mesh (~1 mm) to prevent pests (e.g. insects) from entering the respiration system and disrupting measurements. Unplasticised polyvinyl chloride (UPVC) ducts were connected to the inlet and outlet of the chamber (Supplementary Fig. 2). Cable glands were applied to seal the ports through which the sensor cables passed the duct or

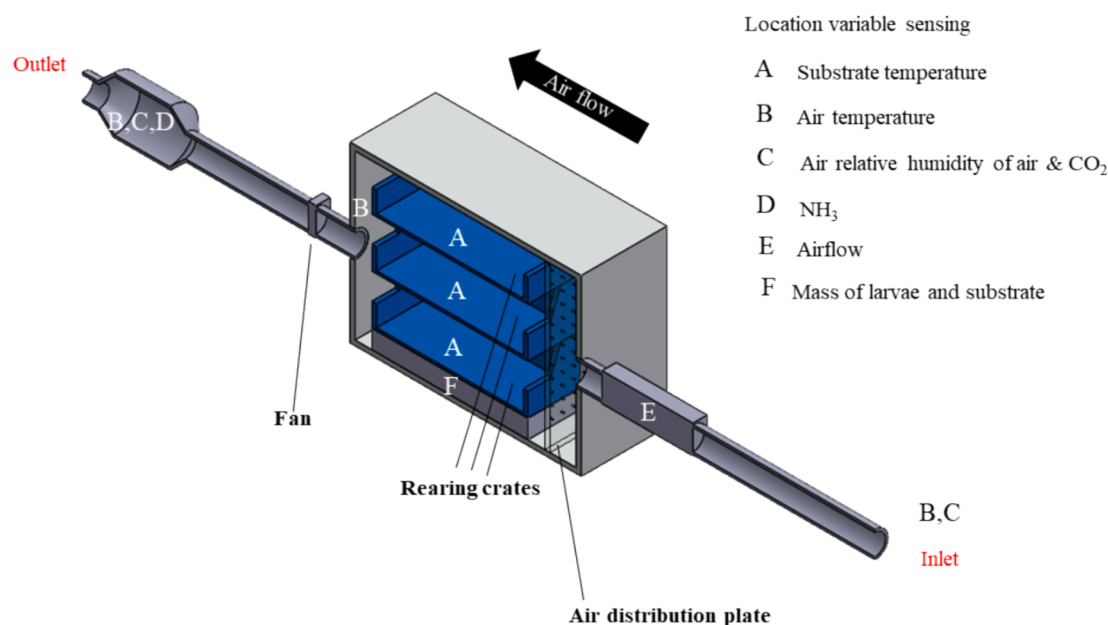


Fig. 1. Lateral cross-section of respiration chamber for fly larvae bioconversion characterisation. Locations of variable sensing, i.e. sensor locations, are indicated using letters.

Table 1
Respiration chamber system specifications.

Parameter	Unit	Value
Interior chamber volume	m ³	0.332
Interior chamber dimensions	[H] mm x [L] mm x [W] mm	630 x 850 x 620
Chamber wall thickness	mm	30
Chamber wall coefficient of thermal insulation	Wm ⁻¹ K ⁻¹	0.023
Air exchange range	m ³ h ⁻¹	18 – 110
Perforated air distribution plate opening area	%	1
Duct diameter	mm	102

chamber wall. The junctions at the interfaces between the chamber walls, cable glands, and ducts were sealed with a multi-use hybrid polymer adhesive.

2.2. Bioconversion and respiration parameters

The BSFL bioconversion process was comprehensively characterised by measuring CO₂ (DOL 119, dol-sensors, Aarhus, Denmark), relative humidity, and temperature (DOL 114, dol-sensors) in the inlet and outlet duct (Fig. 1) using commonly used industrial sensors in livestock farming (Supplementary Table 1 for full sensor list with specifications). The exhaust air was further analysed for NH₃ (DOL 53, dol-sensors). A fan (San Ace 9LG Series, Sanyo Electric Co., Osaka, Japan) in the outlet duct sucked the air through the hypobaric chamber, establishing a mass flow from inlet to outlet. A perforated plate (polycarbonate) was positioned just after the inlet of the chamber to distribute air homogeneously over the inner chamber cross-section (Supplementary Fig. 2, Table 1). An airflow sensor (UltraLink FTCU100, Lindab, Båstad, Sweden) was placed at the end of the inlet duct next to the chamber inlet to reduce turbulence in the airflow and ensure precise flow measurements required for accurate mass balances. To further mimic industrial BSFL bioconversion, three commonly used rearing crates (60 cm × 40 cm × 19 cm, BSF Breeding Box 190 mm, Beekenkamp Verpakkingen, Maas-dijk, Netherlands) were stacked in the chamber, each representing a biological replicate of rearing performance metrics (e.g. larval mass, bioconversion rate, and waste reduction). A scale (FPGA Company,

Tann, Switzerland) consisting of a load cell connected to two aluminium plates (570 mm × 370 mm × 8 mm) as load carriers was placed below the crates to analyse the net mass change of substrate and larvae during the bioconversion. Complementary, three industrial temperature sensors (DOL 12, dol-sensors) per crate recorded the substrate/frass temperature as an indicator of metabolic activity. Each temperature sensor was connected to a weight (50 g) to ensure their submergence and position in the substrate/frass during biowaste bioconversion (corner, intermediate long side, centre; Supplementary Fig. 3). Except for the scale, sensor signals were processed via a controller unit (830 insect monitoring 10⁷ L 32RL, SKOV, Roslev, Denmark) and saved in a CSV file using the corresponding software (Farm Online Export 5.0, SKOV). The fan speed was controlled via a pulse-width-modulated (PWM) signal using a single-board computer (Raspberry Pi 4, Raspberry Pi Foundation, Cambridge, UK).

2.3. Experimental design for technical validation

In the absence of larvae and substrate, the respiration system was validated by completing tests on air tightness, air flow homogeneity and CO₂ recovery. Airtightness is crucial for avoiding drawing in ambient air, which produces underestimated emission results. In addition, homogenous airflow is the key to ensure balanced heat and water transfer throughout the three crates. CO₂ recovery tests are the standard for commercial high-tech respiration chambers for conventional livestock to confirm airtightness, flow rate measurements, and validity of CO₂ measurements. Airtightness was additionally confirmed for all chambers according to Heetkamp et al. (2015) by placing soapy water under maximum airflow at each crucial junction (lid, cable glands, and air ducts) and confirming the absence of bubble formation.

Airflow homogeneity was determined in one of the chambers by measuring air velocity across the chamber cross-section perpendicular to the airflow at five predefined positions over each crate (Supplementary Table 2) with a handheld hotwire anemometer (425, Testo, Titisee-Neustadt, Germany). The anemometer was positioned in the chamber via sealable ports at a constant airflow and 10 s mean air flow velocities were measured for each position.

The CO₂ recovery test was completed in all chambers based on Heetkamp et al. (2015) and compared the difference in CO₂ mass loss from a commercial gas cylinder (99.8 % CO₂) to the CO₂ mass loss

measured in the respiration chamber (Supplementary Equation 1) under constant air flow.

2.4. Experimental design for biological validation and heterogeneous food waste assessment

For the biological validation of the respiration system, i.e. the quality of the setup to collect reproducible metabolic and rearing performance metrics in BSFL feeding trials, the variability of measured parameters within and across respiration chambers was assessed using different rearing substrates. Metabolic metrics included substrate/frass temperature, CO₂ production, and net mass loss of substrate and larvae, whereas the two latter metrics were only measured on chamber level. Rearing performance metrics were based on manually collected data measured on crate level and included bioconversion rate, substrate reduction, survival rate, individual larval mass, total frass yield, total larval yield and frass moisture content. To this end, special focus was placed on the variability of the measured parameters within and across the respiration chambers (Fig. 2) in these trials. First, the variability within the respiration chambers (or across crates in the same chamber) was assessed to validate the performance of crates as biological replicates. As three substrate/frass temperature sensors were placed in each crate, the substrate/frass temperature differences within the crates were assessed. Second, the variability across the respiration chambers was assessed to validate the performance of the respiration chambers as biological replicates. Parameters reported as contents including the final larval mass were aggregated as averages per chamber, whereas the remaining parameters were aggregated as sums per chamber (e.g. average frass moisture content per chamber and sum of larval yield per chamber). Standard deviations thereof were used as measures of variability. However, to render results more comparable across metrics with different units, the coefficient of variation (standard deviation divided by mean) was used for the variables frass yield, total larval yield, water removal, CO₂ production, and mass loss of substrate and larvae.

Biological validation was completed in two separate controlled feeding trials, one with food waste and the other with chicken feed used as control. Each trial involved BSFL bioconversion in all four respiration chambers in parallel. For the two trials, one uniform substrate was used across all four chambers to assess the expected baseline variability of the setup.

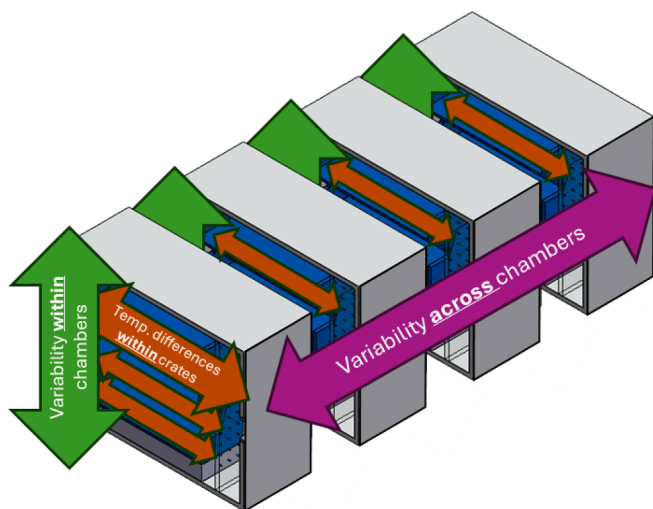


Fig. 2. Schematic overview of the variability assessments for biological chamber validation conducted in the present study. “Temp. differences within crates” refers to the assessment of substrate/frass temperature differences across three different spots in the same crate. The depicted variability assessments were conducted in the two feeding trials using firstly, uniform chicken feed and secondly, uniform food waste across four chambers.

In a third feeding trial on heterogeneous food waste, the composition of the food waste was varied slightly across the four parallel chambers (Supplementary Table 3) to assess the potential of the setup to capture the resulting changes in metabolic processes during BSFL rearing. To simulate a scenario of variation that closely mirrors practical applications, different levels of the common bulking agent cocopeat (0 %, 2 %, 4 % and 6 % of total wet mass) were added to food waste, whereas each chamber received one bulking agent concentration.

2.5. Substrate composition

Food waste consisting mainly of rice, noodles, and, to a smaller extent, meat and vegetables was collected from a university canteen (Ridge View Residential College, National University of Singapore, Singapore), where it was disposed of within approximately 12 h before collection. Following collection, the food waste was stored at 4 °C until use. Before the feeding trial, food waste was ground (Bio Universal Shredder BG2-5.5, voran Maschinen GmbH, Pichl, Austria; see Supplementary Figure 4 for photos). Chicken feed provided by a local distributor was mixed with water to reach a final moisture content of 65 %. Subsequently and before larvae addition, for efficient product separation at harvest, cocopeat was added to the food waste followed by thorough mixing to absorb free water, according to Dortmans et al. (2017), whereas in the trial with heterogeneous substrate across chambers cocopeat was added to induce heterogeneity (Table 2 for cocopeat contents).

2.6. Black soldier fly origin

The BSFL used in the trials originated from a laboratory colony (National University of Singapore) that was maintained for at least 30 generations (Zhang et al., 2024) and initially set up based on wild caught or laboratory cultures. Neonates and larvae of this colony were reared on chicken feed and canteen food waste, respectively. For this study, fly eggs from the colony were placed in an empty container covered with a paper towel for 48 h (Deruytter et al., 2023). Neonates were nursed on chicken feed (62 % water, 9 g of neonates, container: 37 cm × 50 cm × 9 cm). However, for the biological validation with food waste, an external supplier (Ento Industries Pte Ltd., Singapore) provided larvae that had been nursed on a vegetable-based substrate.

2.7. Feeding trials

The three trials were conducted using the described respiration chambers (Table 2 for feeding trial parameters) located in a ventilated laboratory on the National University of Singapore (NUS) campus with a mean relative humidity of 82.8 ± 3.4 % and temperature of 28.5 ± 0.9 °C in 2023, based on one year long measurement from the inlet duct of one chamber. At the beginning of the trial, the substrate was brought to room temperature, thoroughly mixed, and sampled for compositional analysis. Each rearing crate was then gradually filled with the substrate (food waste or chicken feed) on a rotational basis to avoid batch effects. Subsequently, an equal number of 5–7 –day-old larvae were transferred into each crate. Using two spoons, three grab samples, covering the entire surface and vertical substrate/frass profile, were drawn per crate every second day during the trial and at harvest. Samples taken from the same crate (for day 0 and harvest, substrate/frass was mixed instead) were pooled before moisture content, pH, and individual larval mass analyses, whereas material was returned to the respective crate. At harvest, larvae were separated from frass by sieving, whereas if the substrate was too clumpy for sieving larvae were manually picked, and their mass assessed as rearing performance metrics. The total number of larvae per crate at the beginning and at the end of the trial as well as the larval mass were estimated based on random samples of approximately 100 larvae.

Table 2

Parameters of the different feeding trials. The mass of the added bulking agent (coco peat) was not considered for initial feedstock mass. The ventilation rate setpoint sequence indicates the airflow rates that were consecutively set for the specified time periods in all four respiration chambers. The increasing airflow rate ensures that substrate drying is promoted towards the end of a trial to facilitate the separation of substrate and larvae at harvest.

	Unit	Trial		
		Chicken feed	Food waste	Food waste heterogeneous
Substrate type	–	Chicken feed	Food waste	Food waste
Initial larval number	#	12,000	10,800	10,800
Initial wet larval mass	mg	6	13	6
Feeding rate	mg DM larvae ⁻¹ day ⁻¹	25	25	25
Initial feedstock mass	g DM	2100	1890	1890
Trial duration	h	136	155	158
Coco peat addition	% of total wet mass	0	4	0, 2, 4, 6
Ventilation rate setpoint sequence	m ³ h ⁻¹ (hours)	18 (48), 28 (24), 58 (72)	32 (48), 51 (24), 101 (96)	32 (48), 51 (24), 101 (96)

2.8. Substrate, frass and larval characterisation

BSFL bioconversion inputs and outputs i.e. substrates (one sample per substrate before start of the trial), larvae, and frass (one sample per crate at harvest) were characterised for physicochemical properties (Supplementary Table 4 for detailed methods), such as ash, hydrolysed crude fat, and total nitrogen. Nitrogen-to-protein conversion factors of 6.25 and 4.76 were used for substrate and larvae, respectively (Siegrist et al., 2023). Frasses and substrates were analysed for neutral detergent fibre (NDF), acid detergent fibre (ADF), starch, and glucose. Substrate, frass and larvae were analysed for dry matter (DM) by oven drying at 60 °C for at least 3 days or longer until no weight change observed. The pH of the substrate/frass was determined in a 1:5 Milli-Q water suspension using a pH meter (FEP20, Mettler Toledo, Columbus, USA).

2.9. Data post-processing and analyses

Comprehensive BSFL bioconversion process characterisation included sensor-based estimation of CO₂ concentration, NH₃ concentration, airflow, air temperature, air relative humidity, substrate/frass temperature, net mass loss of substrate and larvae, and manually collected data to estimate rearing performance metrics: bioconversion rate, substrate reduction, and survival rate. Sensor-based metabolic process parameters (except for net mass loss of substrate and larvae) were rounded and normalised via the software FarmOnline (SKOV). The data were automatically transferred to cloud storage for post-processing, visualisation, and statistical analysis in R version 4.1.3 (R Core Team, 2022) using RStudio (Posit team, 2023). Post-processing included the removal of erroneous data when the chambers were opened for sample collection based on air flow and CO₂ dynamics. Airflow-based data removal was conducted where differences between two consecutive timepoints exceeded $\pm 5 \text{ m}^3 \text{ h}^{-1}$ because during closed operation air flow differences were much lower than this cut-off. Because these airflow changes had lag effects on the CO₂ concentrations at the point of measurement (outlet duct), removal of erroneous data had to be extended. This outlier removal was based on the difference between actual and smoothed data points based on local regressions (LOESS), where $< 1\%$ of total data was removed.

CO₂ emissions were estimated every two minutes based on the difference in CO₂ between the ingoing and outgoing air multiplied by the current airflow, considering their specific volume at the current temperature (Supplementary Equation 1). Total water removal during the feeding trial was estimated via an input–output based approach i.e. the mass and moisture content of water in the substrate versus in larvae and frass. For a continuous, sensor-based approach of estimating total water removal and substrate/frass dryness, the balance of water entering versus water leaving the chamber via the air was calculated. Briefly, sensor data for temperature and relative humidity were used to calculate the absolute humidity in the ingoing and outgoing air via the respective dew points (Sensirion AG, 2021), specific humidity, and water vapour pressure (Cai, 2019). The absolute water removal per time step and the

total water removal during the trial were determined by factoring in the current airflow rate. The net mass loss of the substrate and larvae in one chamber was calculated based on the difference between the start mass and the sensed mass at each time point during bioconversion. Erroneous mass change data between each measurement (1 min), e.g. due to sampling, were removed ($\geq -0.01 \text{ kg}$ or $\leq 0.01 \text{ kg}$). The average substrate/frass temperature per crate was estimated based on the mean value from the three substrate/frass sensors as an aggregate measure of substrate/frass heat. Here, the temperatures between rearing days 1 (24 h) and 5 (120 h) were used to characterise the substrate/frass temperature during the window of main activity. The rearing performance metrics, bioconversion rate, substrate reduction, and survival rate, were calculated according to Gold et al. (2020) whereas the values for initial substrate mass, frass, and initial and final larval numbers and masses were used. If not indicated otherwise DM values are stated.

3. Results and discussion

BSFL bioconversion, an emerging technology for biowaste valorisation, still has significant process inefficiencies, particularly when treating heterogeneous biowaste streams with varying compositions, such as urban food waste. Therefore, this study proposes and validates a novel respiration chamber system to investigate the impact of varying substrate compositions on BSFL performance parameters that resemble those of industrial insect farming to develop a more reliable and efficient BSFL-based food waste treatment. For the first time, this study presents highly resolved insights into the metabolic processes related to the bioconversion of food waste with BSFL.

3.1. Technical validation

The respiration system was technically validated by measuring air tightness, airflow homogeneity, and CO₂. The results indicate that the chamber performed well in terms of CO₂ recovery. In addition, the chamber exhibited some heterogeneities in the airflow velocity. In the CO₂ recovery test, 95–105 % of injected CO₂ was recorded by the respiration system (Supplementary Table 5). Concerning the airflow homogeneity within the chamber, a trend of higher air velocities in the middle of each crate and in the bottom crate was observed (Supplementary Table 2). However, variabilities of air flow velocities within and across crates were not high (standard deviations: within, $0.03\text{--}0.06 \text{ m s}^{-1}$; across, $0.05\text{--}0.06 \text{ m s}^{-1}$).

One advantage of the presented setup is the possibility of using three stacked industrial-size rearing crates to serve as biological replicates and mimic industrial conditions, in contrast to the open-circuit respiration chamber by Padmanabha and Streif (2019). Yang and Tomberlin (2020) investigated the impact of scale on feeding trials and demonstrated that crate size is crucial for producing data representative of industrial BSFL bioconversion. In contrast, Parodi et al. (2020) had a similar configuration to that of this study; however, the rearing parameters of their three crates were not analysed separately, and airflow velocities over the

crates were not assessed. A technical solution to the measured heterogeneities in the setup used in this study could be the use of lateral fans for air recirculation within the chamber. Additionally, the system of Parodi et al. (2020) had a CO₂ recovery of > 99 % (Heetkamp et al., 2015), and used a CO₂ sensor with a high accuracy, which allowed for the calculation of exact nutrient balances. However, the system proposed in this study significantly improved the accuracy per investment. To the best of the authors' knowledge, the respiration system of Parodi et al. (2020) is of the order of at least one million USD compared to < 60,000 USD for the four-chamber system in this study.

3.2. Biological validation

Biological validation included determining metabolic and rearing performance metrics during controlled feeding trials in the chambers. Comparisons of variability within and between chambers were performed using uniform substrates across the chambers to further validate the applicability of the chamber system for addressing research questions on feeding trials. Within-chamber validation included comparison of rearing performance metrics across three rearing crates in one chamber and the substrate/frass temperature within one crate. The between-chamber validation included a comparison of all measured parameters across the four parallel chambers (Fig. 2).

In general, the results indicated that the presented respiration chambers provide rearing conditions that are well-suited for BSFL rearing, while generating representative results. Metabolic and rearing performance metrics in this study were even higher than those reported previously (Erblund et al., 2021; Gold et al., 2018; Manurung et al., 2016; Naser El Deen et al., 2023; Parodi et al., 2020). Survival rates were around 90 %, final larval mass around 60 mg, bioconversion rates 22–28 %, substrate/frass temperatures > 36 °C for several days, and peak CO₂ production rates between 20–50 g h⁻¹. Peak substrate/frass temperatures of up to 50 °C at individual spots likely did not provide conditions that are optimal for larvae (Chia et al., 2018) but have been reported previously in BSFL and blow fly larvae systems (Erblund et al., 2021; Slone and Gruner, 2007), and proved that the proposed system could be used to investigate the common issue of substrate/frass overheating. These generally high values for larval mass, survival rate and substrate/frass temperature were expected, given that food waste and chicken feed are among the most nutrient- and energy-dense waste streams fed to the BSFL. In fact, the present study emphasizes the significant potential of food waste to be efficiently recycled via BSFL due to its nutritional quality as reported previously (Naser El Deen et al., 2023). For instance, Manurung et al. (2016) reported a final larval mass of only less than 16 mg for BSFL fed with nutrient-poor rice straw. Notably chicken feed and food waste used for the biological validation had similar starch (food waste: 27 %, chicken feed: 30 %) and protein (food waste: 24 %, chicken feed: 24 %) contents (Supplementary Table 3 for more detailed information on substrate characteristics).

3.2.1. Baseline variability within chambers

Rearing performance metrics generally showed small variability across rearing crates, with the frass moisture content as an exception (Supplementary Tables 6 and 7). With both tested substrates (i.e. chicken feed and food waste), variability within chambers was on average 0.5 °C in substrate/frass temperature, < 2 % in bioconversion rate and < 5 mg in final larval mass. Variability in larval survival was lower for food waste (1.6 %) than for chicken feed (7.2 %), which could be due to higher peak substrate/frass temperatures with chicken feed and associated higher larval mortality in some crates. However, for frass moisture content and total frass yield, average variability across crates was larger (5.2 %–9.7 % and coefficients of variation: 17.0 %–19.7 %, respectively). This is likely attributable to recorded airflow heterogeneities (see Section 3.1). Variability of fat and protein content of larvae within chambers was low with < 1 g 100⁻¹ g⁻¹ except for one chamber where protein varied by 1.3 g 100⁻¹ g⁻¹ (Supplementary Table 3). In

frass, the nitrogen content varied by only $\leq 0.12 \text{ g } 100^{-1} \text{ g}^{-1}$. The higher variability of the frass moisture content might have indirectly increased the variability of the other measured parameters; however, if at all, this effect was low, given the low variability in all other parameters. This low variability suggests that individual crates within one chamber are well suited as biological replicates for rearing performance metrics. In fact, by obtaining three biological replicates, one chamber alone would be sufficient to conduct a feeding trial for a comprehensive study of a specific substrate. Depending on the research question for which chambers will be used, a lower variability in frass moisture might still be crucial. Because variability is likely induced by vertical airflow heterogeneities, regular swapping of crate positions could account for airflow heterogeneities and decrease the frass moisture variability.

Additionally, spatial heterogeneities in the substrate/frass temperatures within the crates were assessed. The results from the three temperature sensors in each crate confirmed the importance of using several substrate/frass temperature sensors in each of the industrial-sized crates (Supplementary Figures 3, 5, and 6) for a robust estimation of the average temperature per crate. Hourly average temperatures across sensors differed by more than 3 °C within the same crate over several hours. Differences across sensors within one crate were as high as 6 °C with both substrates. Importantly, at several time points, the temperature trends within a crate developed oppositely, particularly within the first 50 h of the trial (Supplementary Figure 6).

Having robust measurements of substrate/frass temperature is crucial because it is the environment in which larvae live and is, therefore, a main factor for the larval development rate (Wang et al., 2023). In fact, as shown below (Fig. 3A), substrate/frass temperatures can differ between substrates by up to 15 °C. This is remarkable given that the only external input was ambient air; therefore, the substrate/frass temperature was mainly driven by the metabolic activity of larvae and microbes (Li et al., 2023). The observed variability emphasizes the need for future studies to measure substrate temperature. Spatial differences in substrate/frass temperatures are expected in BSFL rearing because of larvae roaming in swarms through the substrate/frass (Shishkov et al., 2019), where areas with denser aggregations of metabolically active larvae likely lead to locally higher substrate/frass temperatures (Gligorescu et al., 2019; Li et al., 2023; Slone and Gruner, 2007). Previous studies conducted daily measurements with an infrared camera to capture the entire spatial variability of the substrate/frass temperature (Li et al., 2023), however only the surface temperatures were considered here. With three continuously measuring substrate/frass temperature sensors per crate, the setup of this study provides a robust estimate. However, further research is necessary to identify the optimal way to capture three-dimensional spatial variability.

3.2.2. Baseline variability across chambers

The variability across the chambers was lower than the previously mentioned variability within the chambers (Fig. 2, Supplementary Table 8). With both substrates, the variability was < 1 % for bioconversion rate, < 0.6 °C for substrate/frass temperature, < 3 mg for final larval mass and 2 % for survival rate. Similarly, for both substrates, the variability (coefficient of variation) was < 2.5 % for larval yield, < 2.7 % for frass yield, < 1 % for water removal, and ≤ 1.9 % for net mass loss of substrate and larvae. Likewise, the variability in product composition was low for fat and protein contents of larvae, varying by 0.55 g 100⁻¹ g⁻¹ and 0.36 g 100⁻¹ g⁻¹, respectively. Total CO₂ production differed moderately between chambers in the feeding trial with food waste with a coefficient of variation of 8 %. However, in the trial with chicken feed, variability of CO₂ production was exceptionally high at 17 %. This outcome might be due to nonuniform waterlogging in chicken feed across crates, resulting in different microbial activities or other unknown reasons (Bekker et al., 2021). The sensor-based approach for water removal estimation yielded higher average values and higher variability than the input-output-based approach, which requires further investigation. However, discrepancies in the average values

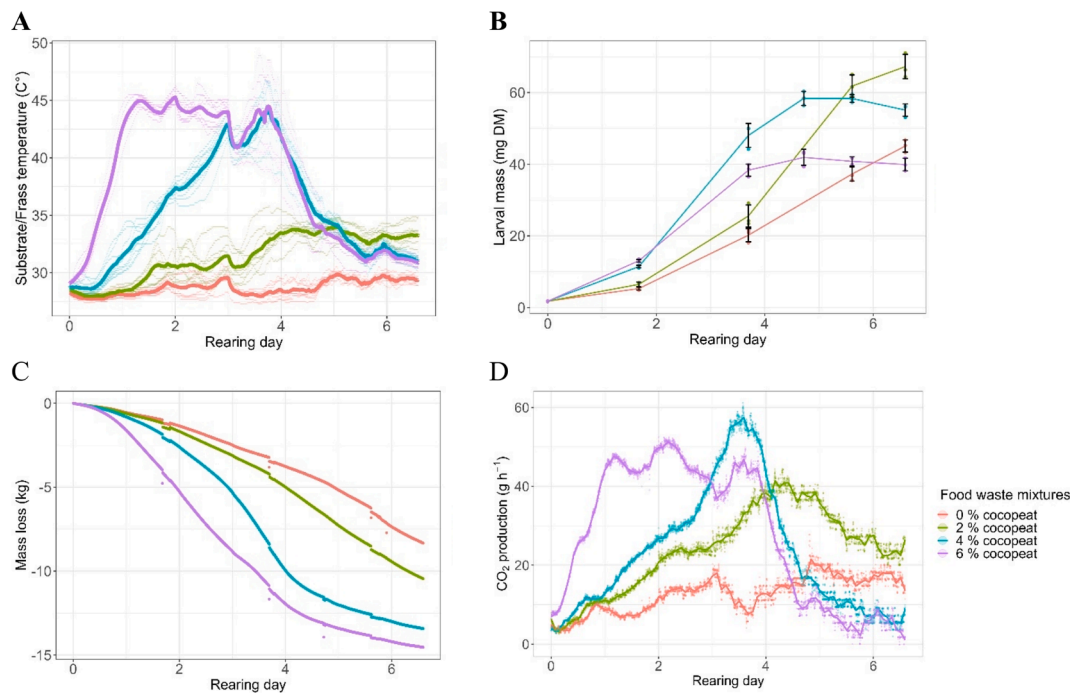


Fig. 3. The time series of metabolic and rearing performance parameters from the bioconversion of food waste with heterogeneous compositions (0 %, 2 %, 4 %, and 6 % fibre addition based on food waste wet mass). A: Substrate/frass temperature ($^{\circ}\text{C}$) data points are shown using semi-transparent dots, and lines show the mean substrate temperature. B: larval mass (mg DM) points representing individual measurements, lines connect mean values ($n = 3$, except day 0) of measurements for each sampling. Error bars indicate the standard deviation around the mean. C: Net mass loss of larvae and substrate in kg. Small breaks in curves arise from chamber openings for sampling. D: CO_2 production rates (measured in g h^{-1}). Transparent points indicate individual values based on sensor measurements. Lines indicate smoothed trend based on individual values via LOESS a non-parametric approach fitting multiple regressions in the local neighborhood with parameter span set to 0.02 indicating the degree of smoothing.

could be partly explained by the release of metabolic water from the substrate, which was not reflected in the initial moisture content. For instance, in this study, approximately 1.4 kg of metabolic water could have been released based on the nutrient composition of food waste and frass and larvae using common conversion factors for metabolic water release (Mellanby, 1942; Morrison, 1953).

Generally, marginal variability across chambers has implications for two possible applications of the respiration system. First, four chambers, each with the same substrate, can be used as replicates to robustly estimate the average total CO_2 emissions, time points of the highest CO_2 production, substrate/frass temperatures, or bioconversion rates when a specific substrate is converted. Second, the chambers can be used to conduct parallel comparisons of the impacts of different substrates. When the chambers are run in parallel, each with a different substrate, the generally determined low across-chamber variability serves as a reference and sets the bar for the effect size. For example, in trials with uniform food waste or chicken feed across the chambers, the variability in the final larval mass was 1 or 2 mg, respectively. Therefore, when comparing the effects of treatments on larval mass across parallel chambers, differences exceeding the 2 mg baseline may be attributed to a treatment effect beyond the variability introduced by the use of different chambers. A baseline of 2 mg is low considering that treatment effects on the final larval mass are generally more pronounced. For instance, NH_3 pretreatments on substrates by Peguero et al. (2023) changed final larval mass by at least 9 mg.

In addition to the provision of aggregated and endpoint values, the respiration chamber system enables investigating and comparing temporal metabolic patterns of the bioconversion process (e.g. substrate/frass and air temperature, relative humidity, gaseous emissions, and net mass loss of substrate and larvae). For instance, CO_2 production curves showed consistent temporal patterns (maxima, minima, and slopes) across chambers for each substrate but altered patterns between different substrates (Supplementary Figures 7 and 8). With food waste,

the CO_2 production rate rose steeply within the first 24 h of the trial, slowed down until day 2, and peaked at a production rate of approximately $50 \text{ g CO}_2 \text{ h}^{-1}$ between days 3 and 4, after which it declined continuously until harvest, intercepted by a short plateau phase towards the end of day 5. In contrast, with chicken feed, the CO_2 production curve was skewed towards the end of the trial, with a slow start, peaking at the end of day 5. Notably the initial larval mass was lower in the chicken feed trial (Table 1) which might have caused a relative delay of the CO_2 peak. Across substrate comparisons show that the system is suitable for highlighting and differentiating the effects of a changing substrate composition on the bioconversion. Continuous time-series data with small sampling intervals, as shown in this study, can be valuable for improving process understanding, but comparative analyses of curves are often qualitative. Future research should explore the feasibility of shape-based quantitative analyses of time series, which could help identify and classify recurring metabolic patterns in bioconversion processes, such as further investigating the contribution of substrate/frass microbes versus BSFL to CO_2 production (Bekker et al., 2021). Potentially suitable methods for quantitative shape based analyses of biological timeseries have recently been described (Dove et al., 2023).

3.3. Substrate/frass temperature, CO_2 production and larval growth with heterogenous food waste

After the biological validation of the setup, a feeding trial with heterogeneous food waste across chambers was conducted to assess the potential of the setup to capture the resulting changes in the metabolic processes in BSFL rearing. The introduction of compositional differences in food waste across chambers led to a strong increase in the variability of bioconversion parameters across chambers compared with trials with uniform substrates across chambers. In fact, the variability across chambers was higher for almost all parameters than the previously

assessed baseline, except for the larval survival rate, which was generally high (Supplementary Table 8). The final larval mass varied by 12 mg larva⁻¹ (coefficient of variation: 23 %). In fact, higher total CO₂ production was accompanied by higher average temperatures in the substrate/frass, higher net mass loss of the substrate and larvae, and faster-growing larvae, especially during the first half of the trial (Fig. 3). In contrast, the bioconversion rate and final larval yield decreased with increasing CO₂ production. Likewise, the four food waste mixtures exhibited distinctive patterns of CO₂ production over time, whereas for the same food waste mixture, the substrate/frass temperature and CO₂ production showed highly similar patterns. Both CO₂ production and heat are indicators of metabolic activity; however, periods of strongest CO₂ production and larval mass growth aligned only for food waste with 2 % and 4 % cocopeat addition. Increased larval or microbial activity might have been a driver of CO₂ and substrate temperature (Bekker et al., 2021; Erbland et al., 2021; Slone and Gruner, 2007). In this study moisture contents differing across food waste mixtures by 3 % might have had an influence on microbial activity and therefore on CO₂ and heat generation in the substrate, however Bekker et al. (2021) indicated that rather differences in substrate moisture contents of 10 % lead to decisive changes in microbial activity. Similarly, strong differences in CO₂ emissions from BSFL bioconversion were observed by Parodi et al. (2020, 2021) notably by using very distinct inputs: an industrial rearing substrate based on yeast concentrate and starch versus pig manure. The peak CO₂ emissions per larva were less than half of those observed in this study. The CO₂ production curves exhibited, depending on the substrate, either one pronounced peak (similar to food waste with 4 % fibre addition in this study) on rearing day 5 of the trial or a gentle, plateau-shaped maximum (similar to food waste with 0 % fibre addition in this study) from rearing days 5 to day 7, respectively. Total CO₂ emissions from pig manure were 40 % lower compared to the industrial rearing substrate, whereas the authors suggested the lower nutritional quality of the latter as an explanation. In this study a nutritional reason for an increased biological activity is unlikely since the common bulking agent, cocopeat, which was used to introduce variation in food waste composition across chambers, had a negligible nutritive value (80 % ADF and 17 % ash content). Additionally, the total amount of food waste per crate did not differ. Notably, past feeding trials frequently used “food waste” as one category when bioconversion parameters or the environmental effects of BSFL rearing on different substrate types were compared (Boakye-Yiadom et al., 2022; Naser El Deen et al., 2023). However, the results of this study suggested that even the bioconversion of nutritionally similar food wastes can lead to process variabilities comparable to those between pig manure and a highly nutritive industrial rearing substrate. In fact, the implications for industrially relevant heat generation and resulting requirements for climate control systems are substantial. Based on the close association between heat and CO₂ in livestock systems (Pedersen and Sällvik, 2002), it can be estimated that during the peak periods in this study, approximately 50 or 20 Watts of heat was generated per crate, depending on the food waste mixture. Extensive heat production requires cooling and therefore energy, which is an environmental but also an economic concern for this waste management technology given that 25 % – 56 % of electrical energy consumption in livestock systems is used for cooling. The findings from this study put into question the way substrates are currently categorised for performance assessments and waste recycling. Further research is necessary to identify the substrate characteristics that lead to the observed temperature patterns, as the management of heat and larval growth are key optimisable variables in the recycling of BSFL biowaste (Li et al., 2023; Wang et al., 2023). The presented time-series data confirmed the potential of the chamber system to characterise and improve the efficiency of food waste bioconversion processes.

4. Conclusions

This study describes and validates a novel, accessible respiration

chamber system that can be used to holistically investigate metabolic processes in BSFL rearing under conditions that resemble those at the industrial scale. Additionally, this study provides unique, highly resolved insights into the metabolic processes that occur during BSFL bioconversion of food waste. Validation trials confirmed the suitability of the setup to address the research questions from a biological and technical perspective. The low baseline variability within and across chambers indicates that crates and chambers could serve as biological replicates to discriminate the impact of different inputs on the bioconversion process. The present study, using the chamber system, showed for the first time that the addition of a nutritionally negligible bulking agent to food waste could lead to processes with strongly differing temporal metabolic characteristics and products. Shown substrate formulation dependent increases in CO₂ emissions are highly relevant for the environmental performance of the BSFL bioconversion. Future work should focus on the technical refinement of the chamber, including the achievement of an accurate estimate of the water balance and the sampling of additional complementary metabolic indicators such as O₂ and CH₄ to compose full energy and material balances. The presented chamber system should be used to further investigate the physical, chemical, and biological characteristics of rearing substrates that drive CO₂ emissions, heat production, and larval growth in BSFL rearing. This will pave the way for better prediction and control of bioconversion processes, which is highly necessary when using heterogeneous biowastes. Additionally, these findings will help to reduce CO₂ and NH₃ emissions from BSFL bioconversion, thereby improving its environmental performance.

CRedit authorship contribution statement

A. Fuhrmann: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **M. Gold:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization. **L.H. Lau Heckmann:** Writing – review & editing, Conceptualization. **P. Pedersen:** Writing – review & editing, Conceptualization. **K. Shakhnovich:** Investigation, Conceptualization. **C.X. Chu:** Investigation. **I. Haberkorn:** Writing – review & editing, Supervision, Project administration. **N. Puniamoorthy:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **A. Mathys:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization, Project administration.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. This research was supported by Better Insect Solutions A/S and SKOV A/S.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.wasman.2024.12.016>.

Data availability

Data and code available at this GitHub repository: https://github.com/Nairdaf/chamber_system_git.git.

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